FINANCIAL MANAGEMENT

Suggested Answers
May 2026

Answer to the Question# 1(a):

Current WACC

Step	Calculation	Result
Cost of Equity (Ke) — CAPM	= 7.5% + 1.40 × 5.0%	= 14.5%
Cost of Debt (Kd) post-tax	= 10% × (1 − 0.225)	= 7.75%
Market Value of Equity (E)	= 100m × BDT 160	= BDT 16,000m
Market Value of Debt (D)	= BDT 4,000m (at par)	= BDT 4,000m
Total Capital (V)	= 16,000 + 4,000	= BDT 20,000m
WACC	**= (14.5% × 16/20) + (7.75% × 4/20)**	**= 11.60% + 1.55% = 13.15%**

Answer to the Question# 1(b):

EPS Analysis
New EBIT = BDT 1,800m × 1.15 = BDT 2,070m
Existing Interest = BDT 4,000m × 10% = BDT 400m
New Shares (Equity option) = BDT 2,000m ÷ (160 × 0.90) = 2,000 ÷ 144 = 13.89 million shares

Item	Equity Option	Debt Option
EBIT	2,070.0	2,070.0
Existing Interest (400)	(400.0)	(400.0)
New Interest (12% × 2,000)	—	(240.0)
EBT	1,670.0	1,430.0
Tax @ 22.5%	(375.8)	(321.8)
EAT	1,294.3	1,108.3
Total Shares	113.89m	100.00m
EPS (BDT)	**11.36**	**11.08**

Answer to the Question# 1(c):

Financial Break-Even (Indifference) EBIT
At the break-even EBIT, both options yield identical EPS. Setting EPS equations equal:

Equity: (EBIT − 400) × 0.775 ÷ 113.89 = Debt: (EBIT − 640) × 0.775 ÷ 100

Step	Workings
Cross-multiply	100 × (EBIT − 400) = 113.89 × (EBIT − 640)
Expand	100·EBIT − 40,000 = 113.89·EBIT − 72,889.6
Solve for EBIT	72,889.6 − 40,000 = 13.89·EBIT → EBIT = 32,889.6 ÷ 13.89
Break-even EBIT	**= BDT 2,367.9 million**

Advice: The projected EBIT (BDT 2,070m) is BELOW the break-even EBIT (BDT 2,368m). Below this threshold, the Equity option yields higher EPS. Management should favour Equity financing if the projected EBIT holds.

Answer to the Question# 1(d):

Sukuk Strategy
Sukuk (Islamic bonds) are asset-backed instruments compliant with Shariah law, which prohibits interest (riba). In Bangladesh's context:

1. **Investor Base:** Sukuk opens the company to Islamic institutional investors (e.g., Islamic banks, takaful funds) who are mandated to avoid conventional debt — substantially widening the investor pool.
2. **Pricing Advantage:** The asset-backed nature of Sukuk may command a lower profit rate vs. unsecured debentures, reducing financing cost.
3. **Regulatory Tailwinds:** Bangladesh Securities and Exchange Commission (BSEC) is actively promoting Islamic capital markets, so regulatory processes for Sukuk issuance are streamlining.

Answer to the Question# 2(a):

(i) Bank Loan
Loan size: BDT 100m
Interest rate: 12%
Tax rate: 30%
Upfront costs: 3m (Application fee: 1% =1m+ Insurance premium: 2% = 2m)

Years 1-2:
Interest = 100m X 12% = 12m
After tax = 12m × (1 - 0.30) = 8.4m per year

Years 3-5:

Year	Opening Balance	Interest @ 12%	After tax	Principal	Total cash outflow
3	100m	12m	8.4m	33.33m	41.73m
4	66.67m	8m	5.6m	33.33m	38.93m
5	33.33m	4m	2.8m	33.33m	36.13m

Year	Cashflows	PV @ 7%	PV @ 15%
0	**97m**	**97m**	**97m**
1	-8.4m	-7.85	-7.31
2	-8.4m	-7.33	-6.35
3	-41.73m	-34.06	-27.45
4	-38.93m	-29.71	-22.27
5	-36.13m	-25.77	-17.96
Total PV		-104.72	-81.34
NPV		-7.72	15.66

Effective cost of financing (IRR) = 7%+ 7.72 / (7.72+15.66) x (15-7)
= 9.64%

(ii) Convertible Bond
Bond size: BDT 100m

Coupon rate: 10%

Maturity: 5 years

Application fee: 0.5% = 0.5m upfront

Tax rate: 30%

Annual coupon: 100m x 10% =10m
After tax = 10m x (1 - 0.30) = 7m per year

Cash flows:
Y0: 100-0.5=99.5m

Years 1-4: -7 per year

Year 5: -7m (coupon) -124m (purchase of shares) =-131m

Year	Cashflows	PV @ 10%	PV @ 12%
0	**99.5m**	**99.5m**	**99.5m**
1	**-7m**	**-6.36m**	-6.25
2	**-7m**	**-5.79m**	-5.58
3	**-7m**	**-5.26m**	-4.98
4	**-7m**	**-4.78m**	-4.45
5	**-131m**	**-81.34**	-74.33
Total PV		-103.53	-95.59
NPV		**-4.03**	3.91

Effective cost of financing (IRR) = 10%+ 4.03 / (4.03+3.91) x (12-10)
= 11.01%

(iii) Right Share Issue

Cost of equity: 15%

Discount: 20% (forces TERP calculation, but for cost we use 15%)

Flotation costs: 3%

Application fee: 0.5%

Underwriting commission: 1%

Total issue expenses: 4.5%

Gross proceeds: BDT 100m
Net proceeds: 100m x (1 - 0.045) = BDT 95.5m

Effective cost of equity:
Required return = 15% on net proceeds.

Effective cost = 15% / (95.5/100) = 15.7% approx.

Answer to the Question# 2(b):

A rights issue does **not automatically dilute control**.
The Chairman can retain control by:

- **Exercising all rights** in proportion to existing shareholding.
- Purchasing **renounced rights** not taken up by other shareholders.
- Encouraging existing controlling shareholders to subscribe fully.

Dilution occurs only if the Chairman does not fully participate.

Answer to the Question# 2(c):

Trade-offs between debt and equity (4 marks)

Cost perspective

- **Debt** is cheaper because interest is tax deductible (after-tax cost 7–8.4%).
- **Equity** is more expensive (15%) and provides no tax shield.

Risk perspective

- **Debt** increases financial risk due to fixed interest and principal repayments.
- **Equity** has no mandatory repayments, improving liquidity and financial flexibility.

Control perspective

- **Debt** preserves existing ownership and voting rights.
- **Equity** may dilute existing shareholders unless they fully subscribe to the rights issue.

Strategic perspective

- **Debt** suits stable cash flows.
- **Equity** strengthens the balance sheet, improves gearing, and may enhance confidence among investors, lenders, and international buyers demanding financial resilience.

Answer to the Question# 3(a):

i) Expected Spot Rate

Range	Midpoint	Probability
10.70 – 10.80	10.75	20%
10.80 – 10.90	10.85	30%
10.90 – 11.00	10.95	30%
11.00 – 11.10	11.05	20%

Expected rate:
=(10.75 x 0.2)+(10.85 x 0.3)+(10.95 x 0.3)+(11.05 x 0.2) = 10.90

BDT receivables: 18,000,000 x 10.90 = Tk. 196,200,000

ii) Forward Contract
Forward bid rate:
10.80−0.20=10.60

BDT receivables: 18,000,000 x 10.60 = Tk. 190,800,000

Answer to the Question# 3(b):

(i) Forward Rate Agreement (FRA)

- Locks in the future interest rate on the investment.
- Protects against falling interest rates.
- No upfront premium.
- Suitable when DAL wants certainty.

(ii) Interest Rate Futures

- Fix future interest rates through exchange-traded contracts.
- Highly liquid and low default risk.
- May not perfectly match DAL's investment amount or maturity (basis risk).

(iii) Interest Rate Option

- Protects against adverse rate movements while allowing DAL to benefit if rates increase.
- Requires payment of an option premium.
- Provides the greatest flexibility.

(iv) Interest Rate Swap

- Exchanges floating and fixed interest payments.
- More suitable for long-term borrowings or investments.
- Less appropriate for a single six-month investment.

Answer to the Question# 3(c):

i) Money Market Hedge
Discount NOK: (18,000,000 / (1+(0.08 x 1) = 16,666,667
Convert at spot bid (10.80): 16,666,667 x 10.80 = 180,000,004 180,000,004
Invest in BDT: 180,000,004 x (1+0.10) = 198,000,004 198,000,004

ii) Option hedge
Worst Case (NOK depreciates)
Use put option at 10.85: 18,000,000 x 10.85 = 195,300,000 198,900,000
Less premium: =250,000
Net = 195,300,000 - 250,000 = 195,050,000

Best Case (NOK appreciates at 11.05)
= 18,000,000 x 11.05 = 198,900,000
Less premium:
Net = 198,900,000 - 250,000 = 198,650,000

Answer to the Question# 4(a):

Portfolio value = BDT 39 million
Contract size = BDT 10 per index point
Exposure in index points = 39,000,000 / 10 = 3,900,000 index points
Current index = 4,200
Contracts required = 3,900,000 / 4,200 = 929 contracts
Premium cost = 929 x 185 = BDT 171,865
Payoff:
If the index falls to 3,850, payoff = (4,200 - 3,850) x 10 x 929 = BDT 3,251,500.
If the index stays above 4,200, payoff = 0

Protective puts guarantee a minimum portfolio value because the strike price sets a floor. If the index falls below 4,200, the option payoff offsets the loss, ensuring liquidation value cannot drop below that level. Proper contract sizing (929 contracts) ensures the hedge exactly matches the portfolio's exposure, making the protection effective and eliminating residual risk.

Answer to the Question# 4(b):

Scenario A: Portfolio = BDT 36 million, Index = 3,850
Unhedged outcome = BDT 36 million
Hedged payoff per contract = (4,200 − 3,850) x 10 = BDT 3,500
Total payoff = 3,500 x 929 = BDT 3,251,500
Gross hedged outcome = 36,000,000 + 3,251,500 = BDT 39.25 million
Net hedged outcome (after premium) = 39,250,000 - 171,865 = BDT 39.08 million

Scenario B: Portfolio = BDT 42 million, Index = 4,350
Unhedged outcome = BDT 42 million
Hedged payoff = 0 (index above strike)
Gross hedged outcome = BDT 42 million
Net hedged outcome (after premium) = 42,000,000 -171,865 = BDT 41.83 million

Answer to the Question# 4(c):

Full or partial hedge?
Full hedge

- Provides maximum protection against a market fall.
- Gives greater certainty of funds for the new plant and reduces the need for costly bridge financing.

Partial hedge

- Costs less and allows greater participation if the market rises.
- However, part of the portfolio remains exposed to market declines, creating a risk of insufficient funds.

Recommendation: Given the requirement to have **at least BDT 38 million** available by 30 June, a **full hedge** is more appropriate because funding certainty is more important than maximizing upside potential.

Answer to the Question# 5(a):

NPV Calculation
Operating cash flows

Year	Sales (Tk.)	Var. cost (Tk.)	Fixed cost (Tk.)	Profit (Tk.)
1	45,000,000	24,000,000	8,500,000	12,500,000
2	53,928,000	29,568,000	9,010,000	15,350,000
3	64,627,315	36,427,776	9,550,600	18,648,939
4	77,449,375	44,879,020	10,123,636	22,446,719
				68,945,658

Note:

- Sales increase at 12% x inflation at 7%
- Var. cost increase at 12% x inflation at 10%
- Fixed cost increase at the inflation rate of 6%

Capital allowances

Year	Tax WDV	Capital Allowances (20%)
0	32,000,000	
1	25,600,000	6,400,000
2	20,480,000	5,120,000
3	16,384,000	4,096,000
4	13,107,200	3,276,800

Tax WDV at the end	13,107,200
Residual value	10,000,000
Capital allowance at end	3,107,200

Taxable profit & tax

Year	Profit (Tk.)	Cap. allow. (Tk.)	Taxable Profit (Tk.)	Tax (Tk.)
1	12,500,000	6,400,000	6,100,000	1,677,500
2	15,350,000	5,120,000	10,230,000	2,813,250
3	18,648,939	4,096,000	14,552,939	4,002,058
4	22,446,719	3,107,200	19,339,519	5,318,368

Working capital

Year	Working capital (15% of sales)
1	6,750,000
2	8,089,200
3	9,694,097
4	11,617,406

Net cash flows

Year	Net cash flows	Remarks
0	(38,750,000)	Investment plus working capital
1	9,483,300	Profit - net working capital - tax
2	10,931,853	Profit - net working capital - tax
3	12,723,572	Profit - net working capital - tax
4	27,128,351	Profit+residula value of the machinery-tax

Net Present Value

Year	Net cash flows	Discount factor @16%	Present Value (Tk.)
0	(38,750,000)	1.000	(38,750,000)
1	9,483,300	0.862	8,174,605
2	10,931,853	0.743	8,122,367
3	12,723,572	0.641	8,155,810
4	27,128,351	0.552	14,974,850
		Net present value:	677,631

Therefore, the project should be accepted as it has a positive NPV.

Answer to the Question# 5(b):

Calculation of the sensitivity of NPV:

Operating cash flows after reduction of 15% sales

	Year 1	Year 2	Year 3	Year 4
Sales revenue	38,250,000	45,838,800	54,933,218	65,831,968
Variable cost	20,400,000	25,132,800	30,963,610	38,147,167
Fixed cost	8,500,000	9,010,000	9,550,600	10,123,636
Profit	9,350,000	11,696,000	14,419,008	17,561,165
Profit as per req (a)	12,500,000	15,350,000	18,648,939	22,446,719
Loss of profit / cash	(3,150,000)	(3,654,000)	(4,229,931)	(4,885,553)
PV Factor @16%	0.862	0.743	0.641	0.552
Loss of profit at PV	(2,715,300)	(2,714,922)	(2,711,386)	(2,696,825)
	(10,838,433)			

The NPV of the proposed scheme would be decreased by Tk. 10.8 million and this means that the amended NPV would be negative. Therefore, the project should not be accepted.

Answer to the Question# 5(c):

Macroeconomic Risks in Bangladesh:

1 High inflation volatility - cost escalation risk
2 Exchange rate risk - imported machinery costs may rise
3 Policy risk - renewable subsidies may change
4 Interest rate instability - this could affect discount rate.

Answer to the Question# 6(a):

Pre merger financials:

DEL:
Net profit = BDT 1,200m
Shares = 200m
EPS = 1,200 / 200 = BDT 6.0
Market capitalization = 200m x 60 = BDT 12,000m
P/E = 12,000 / 1,200 = 10

RPTL:
Net profit = BDT 800m
Shares = 100m
EPS = 800 / 100 = BDT 8.0
Market capitalization = 100m x 120 = BDT 12,000m
P/E = 12,000 / 800 = 15

Post merger share structure:
New shares issued to DEL = 200m x 0.5 = 100m
Total shares = 100m (existing RPTL) + 100m (issued) = 200m

Combined profit including synergies
DEL profit = 1,200m
RPTL profit = 800m
Synergies = 400m
Combined profit = 1,200 + 800 + 400 = BDT 2,400m

EPS before financing = 2,400 / 200 = BDT 12.0

Adjust for bond financing
Green bond = BDT 5,000m
Interest cost = 6% if ESG targets met, 8% if missed
Interest expense = 300m to 400m
Adjusted profit = 2,400 – 300 to 400 = BDT 2,100m to 2,000m
Adjusted EPS = 2,100 / 200 = BDT 10.5 to 2,000/200 =BDT 10.0

Fairness assessment:
DEL contributes 60% of base earnings (1,200 / 2,000) but receives only 50% of equity.
RPTL contributes 40% of earnings yet retains 50% of equity.
Both shareholder groups gain EPS accretion:
DEL: from 6.0 to 10-12.0
RPTL: from 8.0 to 10-12.0

The swap ratio structurally favors RPTL, granting equal control despite lower current profit contribution. Justification rests on RPTL's higher P/E multiple (15 vs 10) and ESG premium. The post-merger EPS is BDT 10-10.5 after synergy and bond financing. While accretive for both sets of shareholders, the 0.5 to 1 swap ratio is not financially fair to DEL on current earnings contribution, as DEL provides the majority of profit but receives only half of equity. A fairer ratio would be closer to 0.6 to 1, unless RPTL's growth and ESG premium are credibly realized.

Answer to the Question# 6(b):

- **Valuation fairness:** The swap ratio should be based on independent valuation and transparent assumptions to ensure equitable treatment of both shareholder groups.
- **Greenwashing risk:** ESG claims must be genuine and supported by measurable performance, not merely used to justify a higher valuation.
- **Shareholder dilution and disclosure:** Issuing new shares dilutes ownership; management must provide complete and transparent information so shareholders can make informed decisions.

Answer to the Question# 6(c):

Benefits

- Aligns employee incentives with long-term ESG objectives.
- Encourages improved environmental performance and sustainable value creation.

Risks

- Unrealistic emission targets may encourage manipulation or discourage employees.
- Excessive share issuance may dilute existing shareholders.

Recommendation

- Set **realistic, independently verified ESG targets**.
- Link rewards to **both financial performance and ESG outcomes**.
- Cap share dilution and disclose the scheme clearly to maintain investor confidence and protect shareholder value.

---The End---